

Gen AI (vs) AI Agents (vs) Agentic AI.

LLM

↳ large language Model.

ChatGPT → GPT Model

Claude → Sonnet
→ Opus
→ Haiku

GitHub Copilot

How LLMs Work?

Ollama → Open
Source
LLMs.

Transformer
Architecture.

Self Attention

How are you?
↓
[.....]

Query (Prompt)



Split the query into tokens



Word Embeddings



Model processes these embeddings



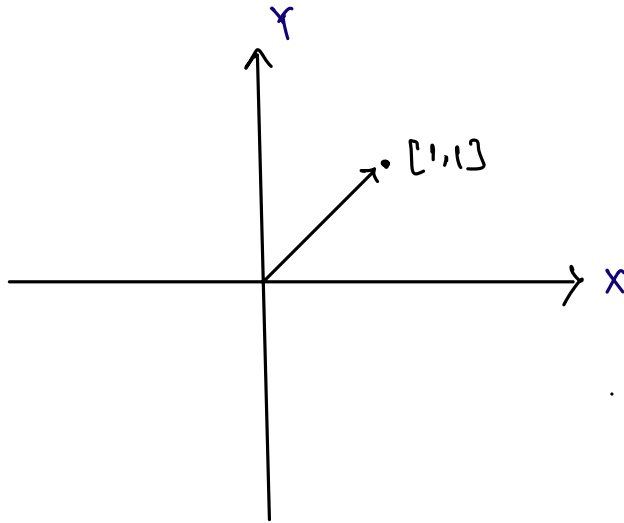
Model predicts the next token
again & again



final response.

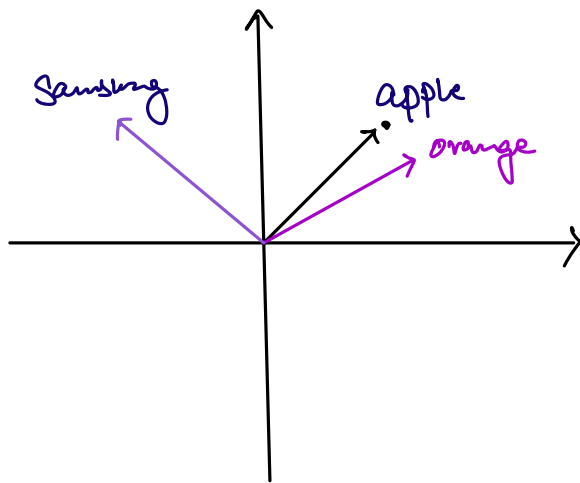
flow $\rightarrow [0.7 \quad 0.1 \quad -0.9 \quad 1.7 \quad 1.4 \quad -1.1 \quad \dots \dots \dots]$

GPT $\rightarrow$ small embedding model $\rightarrow 1536$
 $\rightarrow$ large embedding model $\rightarrow 3072$



Apple : $[0.7 \quad 0.5 \quad -0.1 \quad 0.9 \quad 1.7 \quad 2.5 \quad -2.8 \dots \dots]$

Orange : $[0.6 \quad 0.52 \quad -0.17 \quad 0.97 \quad -5.8 \quad 8.2 \quad 4.9 \dots \dots]$



⇒ LLMs predict the next tokens.

LLM processes text in the form of Tokens.

Token is a small unit of text that is processed by LLM.

unreliable

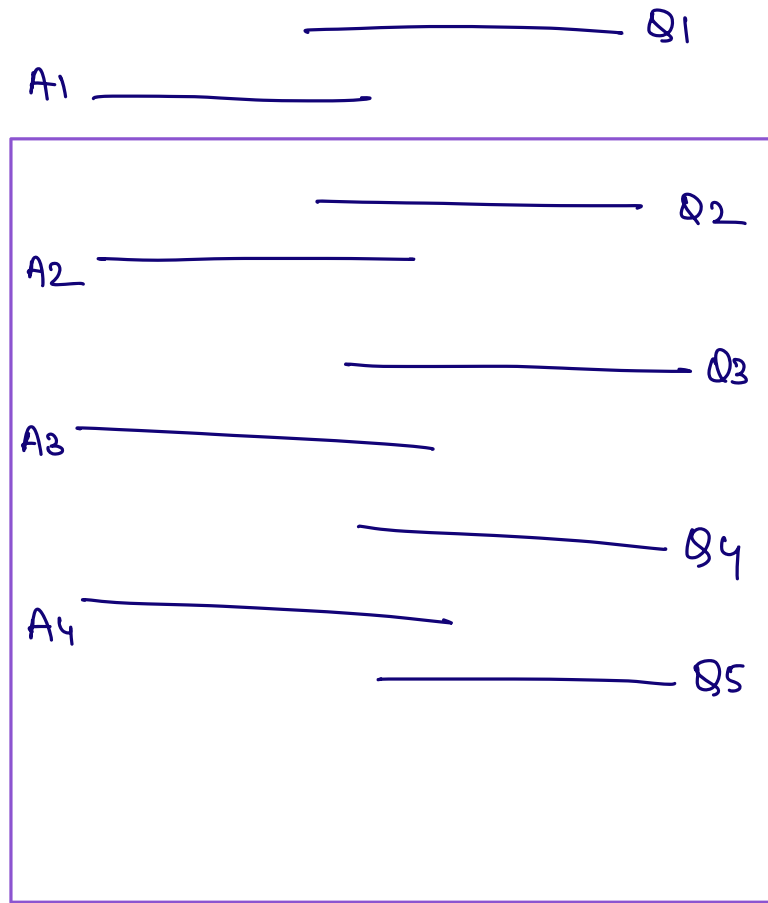
⇒ Tokenization.

LLM doesn't understand the raw text (any language).

Text → Tokens → Vector Embedding → Model processing
↓
Output.

Cost $\propto$ Tokens

Context Window.



Context window is the maximum amount of text an LLM can process at one time.